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Summer Semester 2024 Junior-Prof. Constantin Pohl

"Computational Intelligence" – Exercise 2

Task 1

The IMDb ("Internet Movie Database") dataset contains reviews labeled positive or negative.

a) Load the dataset into your notebook and limit the reviews to 10000 words.
   - How many reviews exist in the dataset for training?
   - How does the first review look like? What does it mean?
   - Is the first review positive or negative?
   - (Optional) You can decode and show the first review by using the word index from IMDb.

b) Create an one-hot encoding for the training and test data.
   - Why is this necessary?
   - How does the first review look like afterward? How long is it?

c) Convert the training and test labels to tensors, i.e. Numpy arrays.

d) Now create a (simple) neural network with dense layers to predict the labels.
   - You can use the Sequential model from Keras (Tensorflow).
   - Start small, define the one-hot encoded format as input, along with suitable activation functions.

e) Compile your model by selecting an optimizer, a loss function, and a metric.
   - For two-class classification, the binary crossentropy is a suitable loss function.

f) Split your training data into training and validation. Start the training procedure with it, 20 epochs could be a good starting point.

g) Plot the resulting loss and accuracy (or your chosen metric). Interpret the results!
   - What happens typically when training too long (i.e. too many epochs)?

h) Change parameters of the training accordingly to g), create a new updated model and train it once more. Evaluate its performance on the test dataset!

Task 2

Reuters is a news agency, classifying news into topics/categories.

a) Load the dataset into your notebook and analyze it the same way as done in Task 1.
    - What is the main difference between both datasets?

b) Apply an one-hot encoding on the data. In addition, you should one-hot encode also the labels.
    - Why was this not necessary in the first task?

c) Create a new neural network for classification and compile it.
    - You can guide yourself by the first task, however, some changes are necessary. Can you identify them?

d) Follow the same procedure. Split your data into training and validation, then start the training procedure. Plot the results on loss and accuracy!

e) You might want to change parameters for a better result. Train the model once more, evaluate on the test data.
    - How well does your model perform? Is it better than random guessing?
    - How do the raw predictions look like, compared to the model from the first task? Can this be explained?